

MAL6150: Probability and Statistics (Autumn 2026–27)
Minor Examination: Questions and Brief Answers

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1. [4 marks]

(a) Let A, B, C be events such that

$$\begin{aligned} P(A) &= 0.42, & P(B) &= 0.38, & P(C) &= 0.30, \\ P(A \cap B) &= 0.17, & P(A \cap C) &= 0.12, & P(B \cap C) &= 0.11, \\ P(A \cap B \cap C) &= 0.05. \end{aligned}$$

Find the probability that *exactly one* of the events A, B, C occurs.

(b) A factory has three machines M_1, M_2, M_3 . They produce 50%, 30%, and 20% of all items, respectively. The probabilities that an item from M_1, M_2, M_3 is defective are 0.01, 0.02, and 0.04, respectively. An item is chosen at random and is found to be defective. Find the probability that it was produced by M_3 .

Brief answer. For (a),

$$\begin{aligned} P(\text{exactly one}) &= P(A) + P(B) + P(C) \\ &\quad - 2\{P(A \cap B) + P(A \cap C) + P(B \cap C)\} + 3P(A \cap B \cap C) \\ &= 1.10 - 2(0.40) + 0.15 = 0.45. \end{aligned}$$

For (b), let D be the event that the item is defective. Then

$$P(M_1) = 0.50, \quad P(M_2) = 0.30, \quad P(M_3) = 0.20,$$

and

$$P(D \mid M_1) = 0.01, \quad P(D \mid M_2) = 0.02, \quad P(D \mid M_3) = 0.04.$$

Thus

$$P(D) = \sum_{i=1}^3 P(D \mid M_i)P(M_i) = 0.019.$$

By Bayes' theorem,

$$P(M_3 \mid D) = \frac{P(D \mid M_3)P(M_3)}{P(D)} = \frac{0.04(0.20)}{0.019} = \frac{8}{19}.$$

2. [4 marks]

(a) Let $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables on the sample space $(\Omega, \mathcal{S})$. Show that

$$\max\{X, Y\}$$

is also a random variable on the same sample space.

(b) Let $(\Omega, \mathcal{S}, P)$ be a probability space, and let $A_1, A_2, A_3, \dots \in \mathcal{S}$. Prove that

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n).$$

Brief answer. For (a), put $Z = \max\{X, Y\}$. For every $t \in \mathbb{R}$,

$$\{Z \leq t\} = \{X \leq t\} \cap \{Y \leq t\} \in \mathcal{S},$$

so Z is a random variable.

For (b), let

$$B_1 = A_1, \quad B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k \quad (n \geq 2).$$

The B_n are pairwise disjoint, $B_n \subseteq A_n$, and $\bigcup_n B_n = \bigcup_n A_n$. Therefore, by countable additivity,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(B_n) \leq \sum_{n=1}^{\infty} P(A_n).$$

3. [5 marks] The function $F : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - \frac{3}{4}e^{-x}, & x \geq 0. \end{cases}$$

(a) Verify that F is a distribution function.

(b) Let X be a random variable with distribution function F . Find

$$P(0 \leq X \leq 2).$$

(c) Is X discrete, continuous, or neither? Give a reason.

Brief answer. For (a), F is nondecreasing and right-continuous, with $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$. For (b), note that

$$F(0^-) = \lim_{t \uparrow 0} F(t) = 0,$$

because $F(t) = 0$ for $t < 0$. Hence

$$P(0 \leq X \leq 2) = F(2) - F(0^-) = 1 - \frac{3}{4e^2}.$$

For (c), however,

$$F(0) = \frac{1}{4}, \quad P(X = 0) = F(0) - F(0^-) = \frac{1}{4}.$$

Thus F has a jump of size $1/4$ at 0 . On $(0, \infty)$, X has density $\frac{3}{4}e^{-x}$, so X is neither discrete nor continuous (in the sense of having a density).

4. [5 marks] Let X be uniformly distributed on $[-1, 2]$, and define

$$Y = X^2.$$

Find the distribution function of Y for all $y \in \mathbb{R}$. Then find its density.

Brief answer. Since X has density $1/3$ on $[-1, 2]$, we compute F_Y from

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y).$$

For $0 \leq y < 1$,

$$\{X^2 \leq y\} = \{-\sqrt{y} \leq X \leq \sqrt{y}\},$$

which is an interval of length $2\sqrt{y}$ inside $[-1, 2]$. Thus $F_Y(y) = 2\sqrt{y}/3$. For $1 \leq y < 4$, the lower endpoint $-\sqrt{y}$ lies to the left of the support, so

$$\{X^2 \leq y\} \cap [-1, 2] = [-1, \sqrt{y}],$$

which has length $1 + \sqrt{y}$. Therefore

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \frac{2\sqrt{y}}{3}, & 0 \leq y < 1, \\ \frac{1 + \sqrt{y}}{3}, & 1 \leq y < 4, \\ 1, & y \geq 4. \end{cases}$$

Differentiating on the intervals where F_Y is differentiable gives the density

$$f_Y(y) = \begin{cases} \frac{1}{3\sqrt[3]{y}}, & 0 < y < 1, \\ \frac{1}{6\sqrt{y}}, & 1 < y < 4, \\ 0, & \text{otherwise.} \end{cases}$$

5. [4 marks] Let X be a non-negative integer-valued random variable, that is,

$$X(\omega) \in \{0, 1, 2, \dots\} \quad \text{for all } \omega \in \Omega.$$

Prove that if $E(X) < \infty$, then

$$E(X) = \sum_{n=1}^{\infty} P(X \geq n).$$

Brief answer. Writing $p_k = P(X = k)$ and using only nonnegative terms,

$$\begin{aligned} E(X) &= \sum_{k=1}^{\infty} k p_k = \sum_{k=1}^{\infty} \sum_{n=1}^k p_k \\ &= \sum_{n=1}^{\infty} \sum_{k=n}^{\infty} p_k = \sum_{n=1}^{\infty} P(X \geq n). \end{aligned}$$

6. [3 marks] Let X be a random variable such that

$$E(|X|^3) < \infty.$$

Prove that

$$E(|2X^3 - 3X^2 + 5X - 7|) < \infty.$$

Brief answer. Since $E(|X|^3) < \infty$, the theorem on lower-order moments gives

$$E(|X|^2) < \infty \quad \text{and} \quad E(|X|) < \infty.$$

Also, by the triangle inequality,

$$|2X^3 - 3X^2 + 5X - 7| \leq 2|X|^3 + 3|X|^2 + 5|X| + 7.$$

Taking expectations, the right-hand side has finite expectation. Hence

$$E(|2X^3 - 3X^2 + 5X - 7|) < \infty.$$

Bonus question. [4 marks] Let $A_1, A_2, \dots$ be events such that

$$\sum_{n=1}^{\infty} P(A_n) < \infty.$$

Prove that

$$P\left(\limsup_{n \rightarrow \infty} A_n\right) = 0.$$

Brief answer. Let

$$B_n = \bigcup_{k=n}^{\infty} A_k.$$

Then $B_n \downarrow \limsup_{n \rightarrow \infty} A_n$. By countable subadditivity,

$$P(B_n) \leq \sum_{k=n}^{\infty} P(A_k) \rightarrow 0.$$

By continuity from above,

$$P\left(\limsup_{n \rightarrow \infty} A_n\right) = \lim_{n \rightarrow \infty} P(B_n) = 0.$$